Saatwik Sairaam Vasamsetti

Data Analyst · Data Engineer

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PROFILE

Data-focused engineer who ships production-shaped systems and measures them. Recent work spans Trade Research, where Python computes every figure and a verifier traces each one in a generated report back to its source data; the backend and cloud database of an offline-first emergency platform whose TypeScript/PostGIS API passes 1,603 assertions across six suites; and a privacy engine at Qaulium AI covered by 360 unit and integration tests. Comfortable with relational schema and index design, SQL, transactional concurrency, Python data pipelines, model evaluation and BI — and in the habit of reporting the result even when the result is negative.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, SQL, Java, HTML, CSS
Backend & APIs: Node.js, Fastify, Express, FastAPI, REST design, JWT auth with rotating refresh tokens, RBAC, Zod validation, webhooks
Frontend: React 19, Next.js 16, Angular, Tailwind CSS, Jetpack Compose, progressive web apps, dependency-free canvas UI, accessibility (A11y)
Databases: PostgreSQL 16 + PostGIS, MySQL, Drizzle, Prisma, schema & index design, transactional concurrency control
Data & ML: pandas, NumPy, scikit-learn, Ultralytics YOLO11, ONNX, exploratory data analysis, walk-forward back-testing, conformal prediction, Monte Carlo simulation
Visualisation & BI: Power BI, Tableau, Matplotlib, Chart.js, Advanced Excel dashboards, data storytelling
Cloud, Tooling & QA: AWS, Docker & Compose, Git, GitHub Actions CI, Gradle, Vitest, ESLint, PyInstaller

EXPERIENCE

Full Stack Web Developer Intern — Qaulium AI Private Limited *Jul 2026 – Present*

- Built the privacy engine of the Qualium Quantum Browser in TypeScript — ad and tracker blocking, cookie and storage isolation, fingerprint, WebRTC and DNS protection — as pure decision functions with no I/O, so every rule is testable without a browser.
- Covered it with 360 Vitest unit and integration tests across 22 files and 24 Playwright end-to-end tests, gated on typecheck and lint.

Co-Founder — PathAhead, career-guidance platform *Jun 2026 – Present*

- Built the live platform: 12 career roadmaps, a Career Builder and an AI counsellor grounded in PathAhead's own content, on two Netlify functions.

Campus Ambassador — SmartED Innovations *Jun – Jul 2026*

- Led campus outreach for SmartED's initiatives; formally recognised for leadership, dedication and the connections built across the student community.

Data Analytics Job Simulations — Deloitte, Quantium & GenAI programmes via Forage *Jun – Aug 2026*

- Deloitte (Jul 2026): completed practical tasks in data analysis and forensic technology.
- Quantium (Aug 2026): data preparation and customer analytics, experimentation and uplift testing, and the commercial application of the results.
- GenAI-Powered Data Analytics (Jun 2026): exploratory data analysis and risk profiling on a lending dataset, delinquency prediction, and the data storytelling behind an AI-driven collections strategy.

SELECTED PROJECTS

Trade Research — Verifiable Research & Execution Platform *Aug – Sep 2026*

Stack: *Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller*

- Built a leakage-free walk-forward back-test benchmarking six scikit-learn models against naive and drift baselines, with conformal prediction intervals reporting honest held-out coverage, Monte Carlo price paths, and a cost-aware trading simulation reporting Sharpe and drawdown.
- Inverted the usual "AI analyst" design so Python computes and the model only interprets: indicators are calculated in code, financials come from SEC EDGAR XBRL with accession numbers, missing data stays null, and a verifier machine-checks every figure in the finished note back to the snapshot.
- Implemented the execution platform end to end — FastAPI, PostgreSQL and React wiring alerts through risk checks and an OMS to the broker — plus an autonomous MetaTrader 5 demo harness with scheduled wind-down.

- Reported the negative result rather than burying it: none of the strategies measured here has a demonstrable edge, and the README says so first.
- Measured the packaging decision instead of preferring one: PyInstaller onedir starts in 1,000 ms against onefile's 5,400 ms, so the 24-command toolkit ships as a directory.

Private repository — available on request

RoadAssist Bharat — Offline-First Emergency Mobility Platform *Aug – Oct 2026*

Stack: *TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, zod, Docker, GitHub Actions*

- Led the backend & cloud-database workstream of a four-person team: an 86-route Fastify modular monolith over a 60-table PostGIS schema with 151 indexes and 13 migrations, serving four browser surfaces, a native Android client and a feature-phone SMS channel.
- Platform verified by 1,603 assertions across six suites, zero failures — including 144 security attacks that must each be refused (cross-tenant access, role escalation, SQL injection, forged and alg:none tokens, unsigned webhooks) and 112 concurrency tests.
- Settled every race in the database rather than by timing: SELECT ... FOR UPDATE on booking rows with offer expiry re-checked under the lock, a partial unique index making "paid once" true under simultaneous confirmations, and a hash-chained audit log that Postgres RULES make immutable.
- Implemented auth end to end: hashed SMS one-time codes, 10-minute access tokens and rotating refresh tokens whose reuse revokes the whole family, with zod validation on every route.
- The platform's health endpoint distinguishes "no network" from "platform unwell": CI kills PostgreSQL under a running API and checks /health returns 503 while /v1/ping stays up, then rehearses a backup and restore and re-verifies the audit chain.

github.com/saatwik-1157/roadassist-bharat

MachineSense AI — Explainable Industrial Intelligence *Smart India Hackathon 2026*

Stack: *Python, FastAPI, JavaScript, Canvas*

- Built an Industry 4.0 prototype for Indian industrial units: a FastAPI backend simulating a fleet of IoT-instrumented machines (vibration, temperature, current, sound, power) behind an analytics engine and a six-view real-time dashboard.
- Implemented the AI layer — a weighted 0-100 machine health score, MAD z-score anomaly detection, remaining-useful-life and failure-probability forecasting, and energy/cost/CO2 optimisation across the fleet.
- Made every score explainable: each health number is decomposed by sensor with plain-language root-cause reasoning, and a digital twin simulates a maintenance action before anyone touches the real machine.
- Mirrored the same scoring logic in dependency-free browser JavaScript so the demo behaves identically with the backend down, and labelled on screen which mode is running.

github.com/saatwik-1157/machinesense-ai

Smart University Management System (SUMS) *2026*

Stack: *MySQL, JavaScript, Chart.js, CSS3*

- Designed a normalised MySQL schema for an enterprise university system managing students, courses, faculty and academic records, with the query layer behind it.
- Built a glassmorphism front end with Chart.js dashboards surfacing enrolment and performance insights over the live data.

github.com/saatwik-1157/database_management_SUMS

OfferPilot — Job-Application Platform for F-1 / OPT Students *2026*

Stack: *JavaScript, HTML, CSS, localStorage*

- Built a dependency-free static platform — marketing site, onboarding, client dashboard, profile, legal pages and a passcode-gated ops console — with no build step and full localStorage persistence.
- Shipped the RezForge resume-tailoring engine, an interview kanban, a referral programme and analytics charts on a hand-written SVG engine, plus an OPT Runway Calculator computing work-authorisation runway, unemployment-day budget and H-1B lottery timing.
- Labelled the prototype honestly on the page: the match and resume engines are deterministic mocks and no real applications are submitted.

github.com/saatwik-1157/offerpilot

Angular Applications — ShopCart, Notes, Search & Auth *2026*

Stack: *Angular 22, TypeScript 6, standalone components, signals, Vitest*

- **ShopCart (e-commerce):** searchable, category-filtered catalogue, product detail pages, a quantity-aware cart with computed totals, and signup/sign-in with a route guard protecting checkout — 14 Vitest unit tests across cart, accounts and header.
- **Search template:** a user list loaded from JSON through a service and filtered on screen as the query is typed, matching on either name or email address.
- **Notes app:** add, update and delete notes held in a service-owned signal, rendered through separate form, list and single-note components.

- **Authentication app:** a login page validated in a service, with the home page reachable only through a route guard after a successful sign-in.
- Every experiment's lab record is generated from the live source on disk rather than written by hand, so the submitted document cannot disagree with the code that ran.

ExamPlanner — Android Study Planner *2026*

Stack: *Kotlin, Android, Java*

- Built a native Android study planner with authentication, exam and schedule management, and a Pomodoro timer with customisable focus and break intervals.

GenAI-Powered Data Analytics — Job Simulation *Forage · Jun 2026*

Stack: *Python, pandas, GenAI, data storytelling*

- Performed exploratory data analysis and risk profiling on a real-world lending dataset, built a model to predict customer delinquency, and delivered the data storytelling behind an AI-driven collections strategy.

PathAhead — Career-Guidance Platform *Jun 2026 – Present*

Stack: *HTML, CSS, JavaScript, Netlify Functions*

- Built the live platform (pathahead22.netlify.app): 12 career roadmaps with downloadable PDFs, a Career Builder that turns branch, year and interest into a plan, and an AI counsellor grounded only in PathAhead's own content — 45 commits.
- Static site with two Netlify serverless functions (counsellor, leads), so it stays fast on the patchy mobile data most students open it on.
- Shipped the surrounding promotional kit alongside it: presentation, brochure and tri-fold materials, so the launch did not depend on a separate design pass.

github.com/saatwik-1157/project_PathAhead

Additional projects

- Gemini Chat Client — a complete browser-only chat client: multi-thread sidebar, four personas, vision and voice input, Markdown rendering sanitised with DOMPurify, localStorage persistence, no server and no build step.

CERTIFICATIONS

- Google Analytics Certification — Google, May 2026 – May 2027 (ID 183723043)
- GenAI-Powered Data Analytics (Job Simulation) — Forage, Jun 2026
- Data Analytics Job Simulation — Deloitte via Forage, Jul 2026
- Data Analytics Job Simulation — Quantium via Forage, Aug 2026
- Introduction to AWS — AWS Training & Certification, May 2026
- AWS Certified Cloud Practitioner (CLF-C02) Domain 1 Review — AWS, May 2026
- AWS Certified Data Engineer – Associate (DEA-C01) Exam Prep — AWS, May 2026
- Exploring SAP Analytics Cloud — SAP Learning, Jun 2026
- Becoming an SAP Data Architect — SAP Learning, Jun 2026
- Architecting Agentic AI Business Solutions — Microsoft Learn badge, 2026

ACHIEVEMENTS

- Led backend & cloud database for RoadAssist Bharat, a four-person platform verified by 1,603 assertions across six suites with zero failures.
- Earned ten professional certifications across data analytics and cloud — AWS, Google and SAP and Microsoft — including three Forage job simulations (Deloitte, Quantium, GenAI analytics).
- Attended the IIT Hyderabad Student Body Workshop on Career, AI & Campus Experience, Jun 2026.
- Participant, HackDevengers 1.0 hackathon — organised by Devengers, 2026.
- Scored 94.8% in Class XII and 81.17% in Class X; recognised as Campus Ambassador, SmartED (2026).
- Designed, built and deployed a dependency-free personal portfolio at saatwik-1157.github.io.

EDUCATION

VIT-AP University, Amaravati *2024 – 2029*

Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa) *CGPA: 7.55 / 10*

Coursework: *Advanced Programming, Data Structures & Algorithms, Database Management Systems, Software Engineering, Frontend Design & Testing*

SR Junior College *2022 – 2024*

Intermediate (Class XII, MPC) *94.8%*

DAV Public School

Secondary School (Class X) *81.17%*